Modelling Drug-Induced Changes in Bursting Dynamics of the *Aplysia R15* Cell

MATH3888: Projects in Mathematics (Stream 1) — Group 13

Group Member	**SID**	**Slide**
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Calvin Lai	530598916	Slide 3 (Baseline Dynamics)
Nicholas Bradshaw	540701546	Slide 4 (Slow Variable Analysis)
Peng Annan	540046698	Slide 5 (Ouabain Analysis)
Lucie Roche	520449389	Slide 6 (TTX Analysis)
Theodore Huxtable	540758539	Slide 7 (TTX + Ouabain Analysis)
Emma Karpes	540695186	Slide 8 (Conclusion & Limitations)

Physiology and Master Equations (Perry Cahill — SID: 540744381)

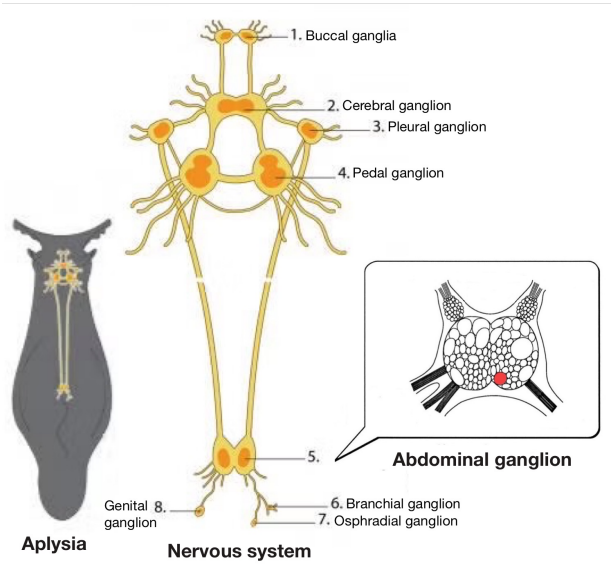


Figure: Chiu and Zare [1998]

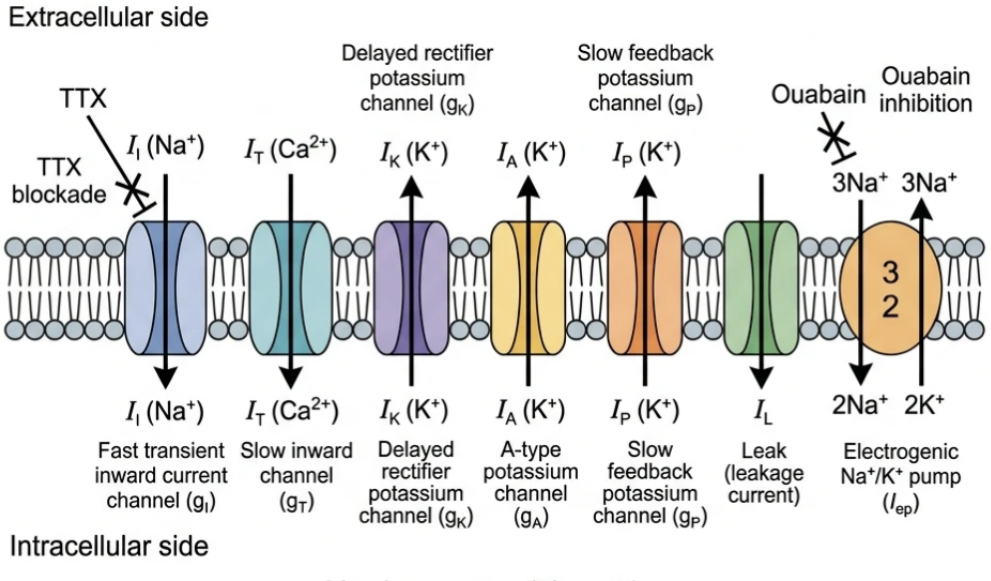


Figure: Ion channels

Membrane Potential

$$
\begin{aligned}
C\dot{V} = &(g_I X_I^3 Y_I + g_T)(V_I - V) \\
&+ (g_K X_K^4 + g_A X_A Y_A + g_P X_P)(V_K - V) \\
&+ g_L(V_L - V) \\
&+ I_{\text{ep}} + I_{\text{ext}},
\end{aligned}
$$

6 Gating Equations

$$
\dot{X}_I = \frac{S_I(V) - X_I}{\tau_{X_I}(V)}, \qquad \dot{Y}_I = \frac{Z_I(V) - Y_I}{\tau_{Y_I}(V)},
$$

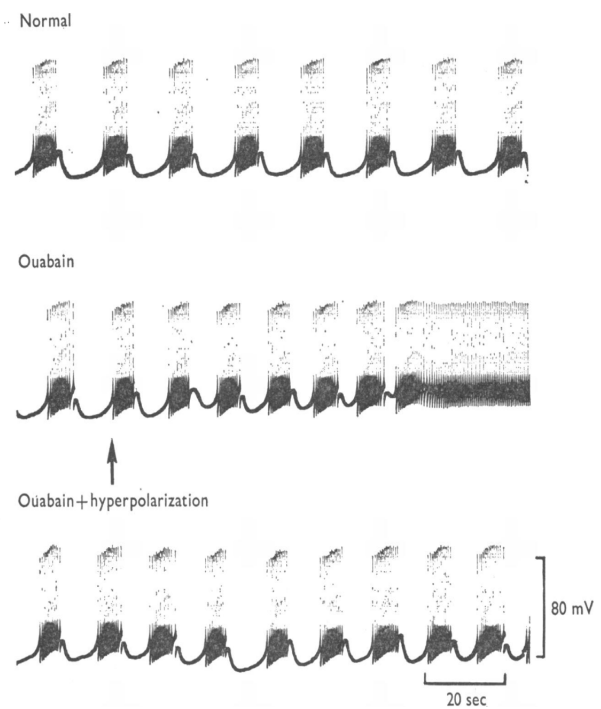
$$
\dot{X}_K = \frac{S_K(V) - X_K}{\tau_{X_K}(V)}, \qquad \dot{X}_A = \frac{S_A(V) - X_A}{\tau_{X_A}},
$$

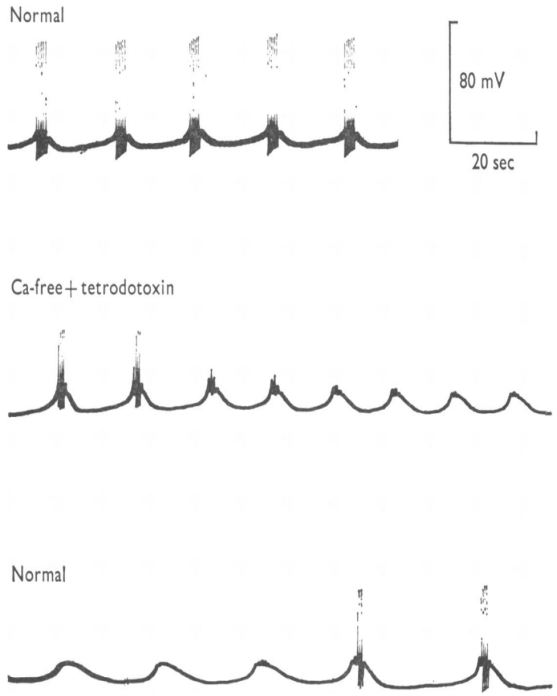
$$
\dot{Y}_A = \frac{Z_A(V) - Y_A}{\tau_{Y_A}}, \qquad \dot{X}_P = \frac{S_P(V) - X_P}{\tau_{X_P}}.
$$

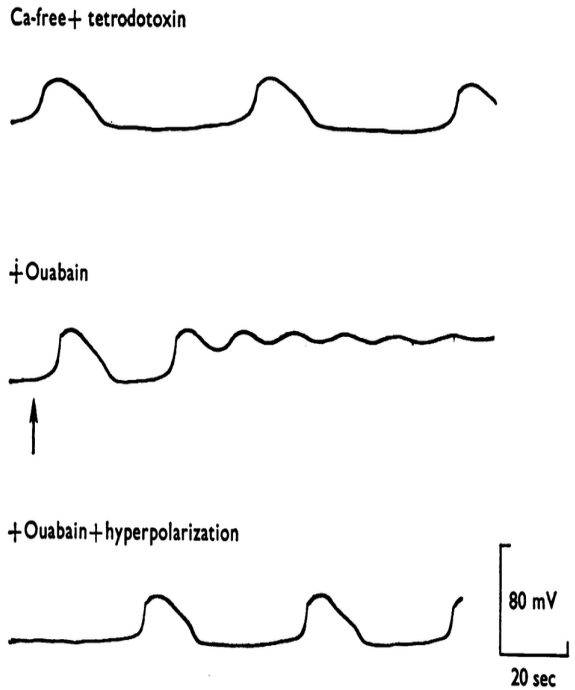
Experimental Protocols (Ashley Chen — SID: 530017071)

Intracellular membrane-potential recordings from the R15 neuron of *Aplysia californica* under drug treatment and current injection, from Junge and Stephens [1973].

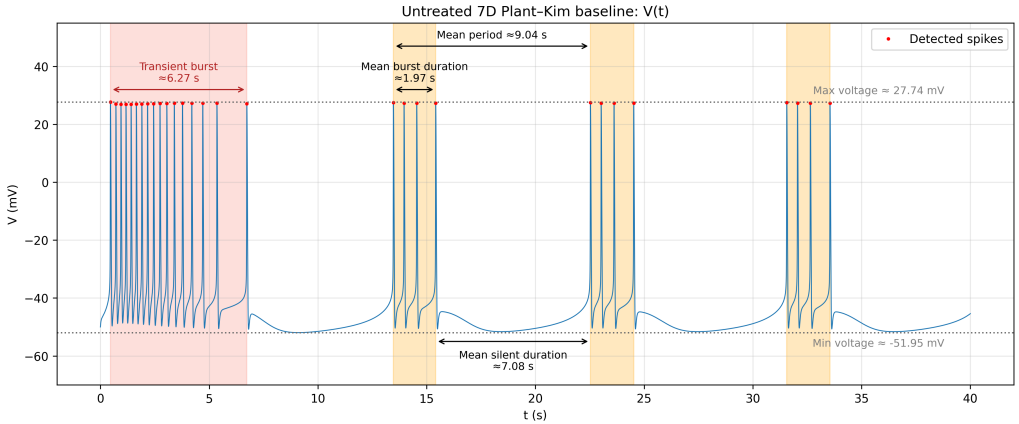
Ouabain	**TTX**	**TTX + Ouabain**
0.4 mM Inhibits the Na^+/K^+ pump	25 μg/mL, Ca^{2+}-free saline Blocks the TTX-sensitive transient inward current	25 μg/mL TTX, Ca^{2+}-free saline + 0.4 mM ouabain



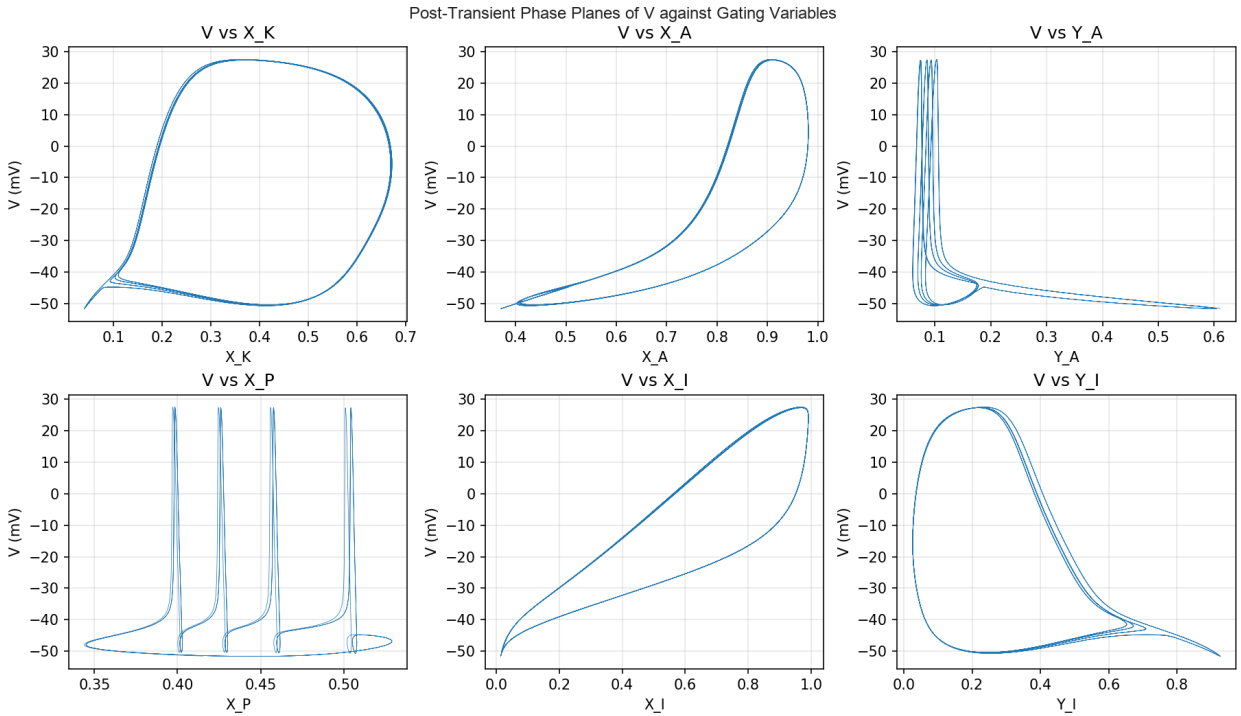




Baseline Dynamics (Calvin Lai — SID: 530598916)

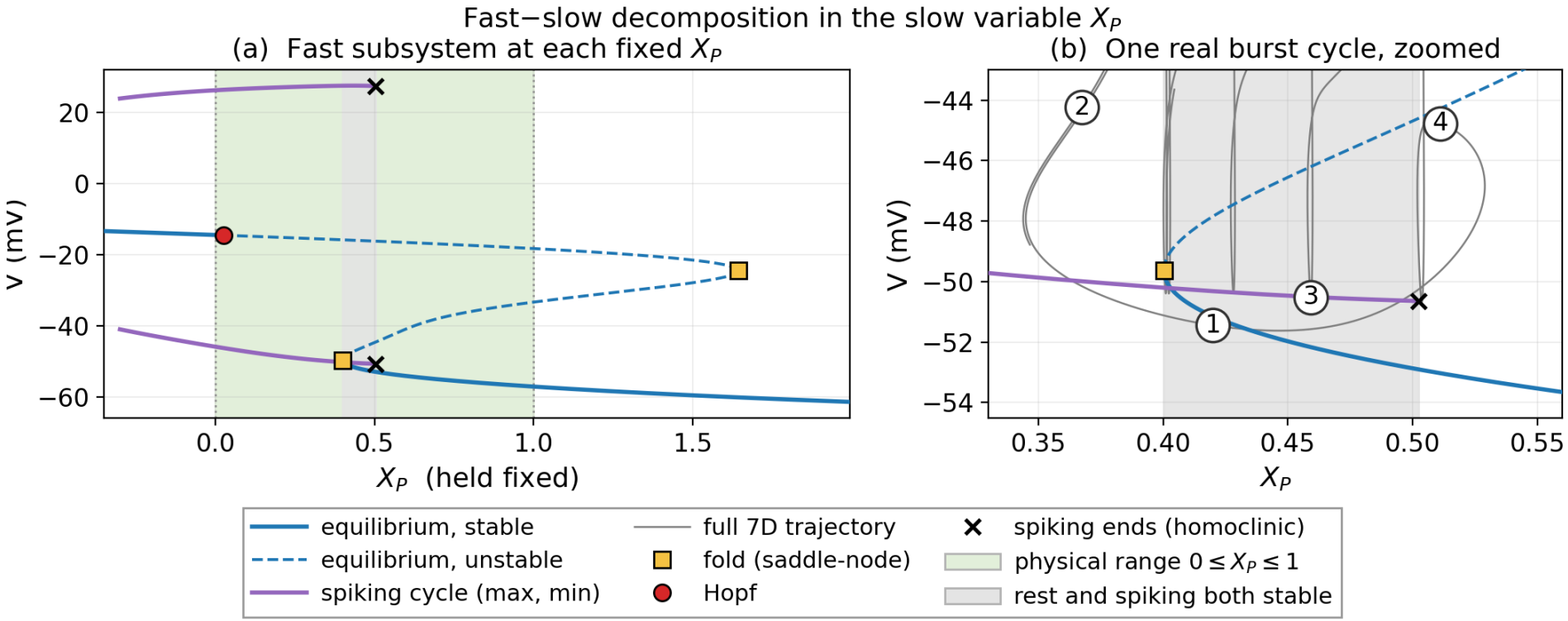


(a) Time trace of membrane voltage V under the baseline model; bursting behaviour shown and durations numerically computed.



(b) Phase-plane projections of membrane voltage V against the six gating variables. The slow variable X_P evolves alongside rapid voltage excursions.

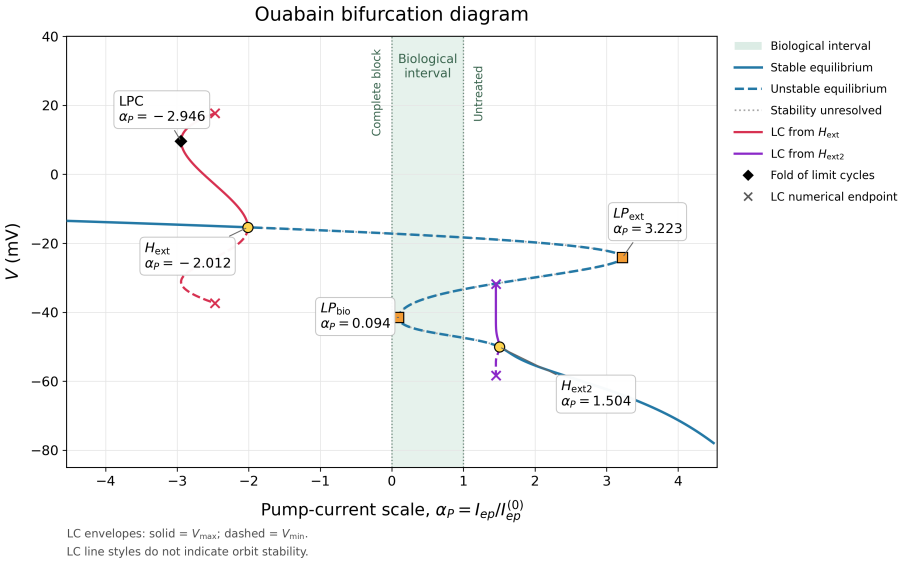
Slow Variable Analysis (Nicholas Bradshaw — SID: 540701546)



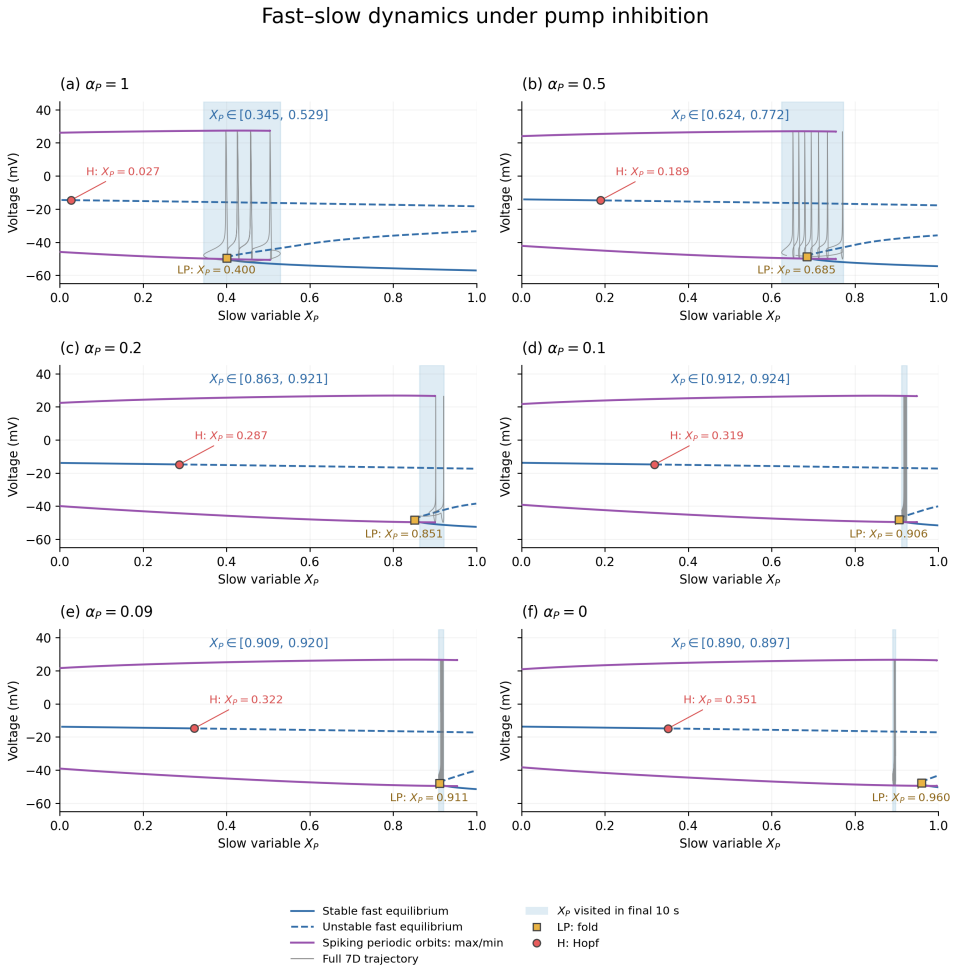
Burst boundaries $X_P = 0.4003$ (fold) and 0.5036 (homoclinic), computed **without simulating a burst**.
Observed last spike: $X_P = 0.502$.

Bifurcation Analysis: Ouabain (Peng Annan — SID: 540046698)

Investigating the change from bursting to continuous spiking as pump inhibition increases ($\alpha_P = I_{ep}/I_{ep}^{(0)}$).



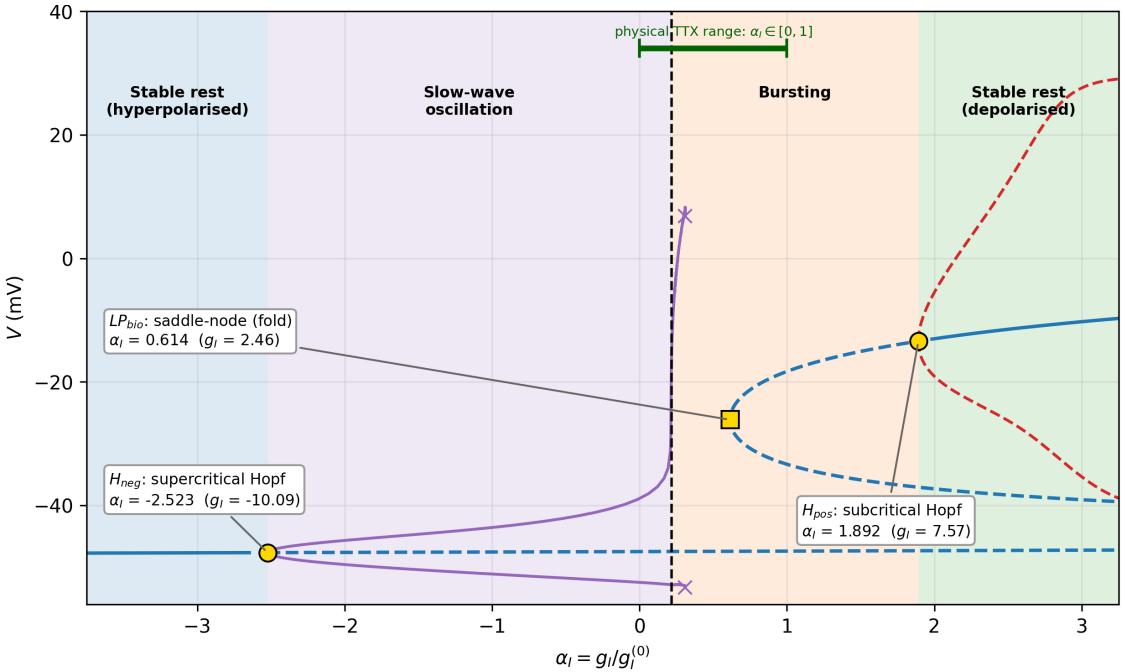
Equilibrium bifurcation diagram

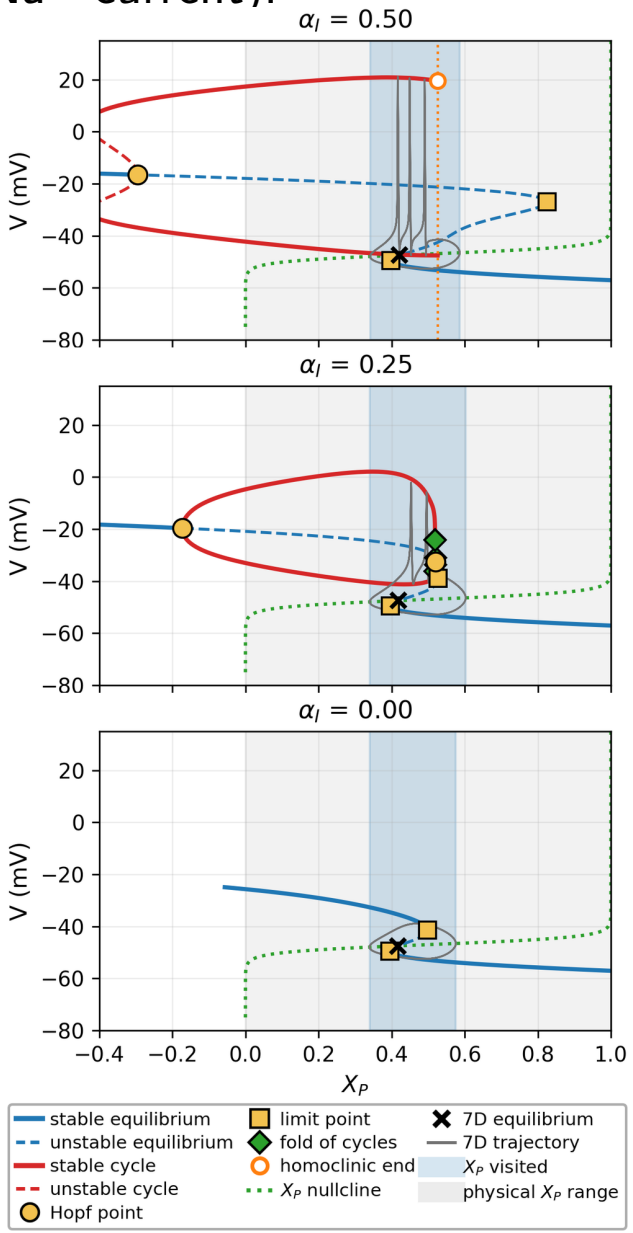


Fast–slow comparison

Bifurcation Analysis: TTX (Lucie Roche — SID: 520449389)

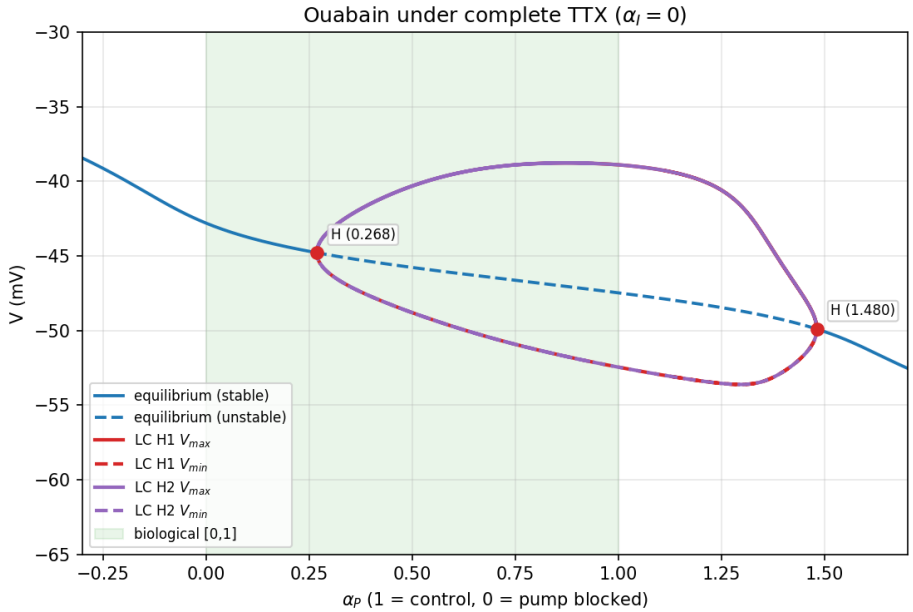
Investigating the bursting to oscillatory pattern observed with increased TTX ($\alpha_I = g_I / g_I^{(0)}$, fast transient inward Na^+ current).



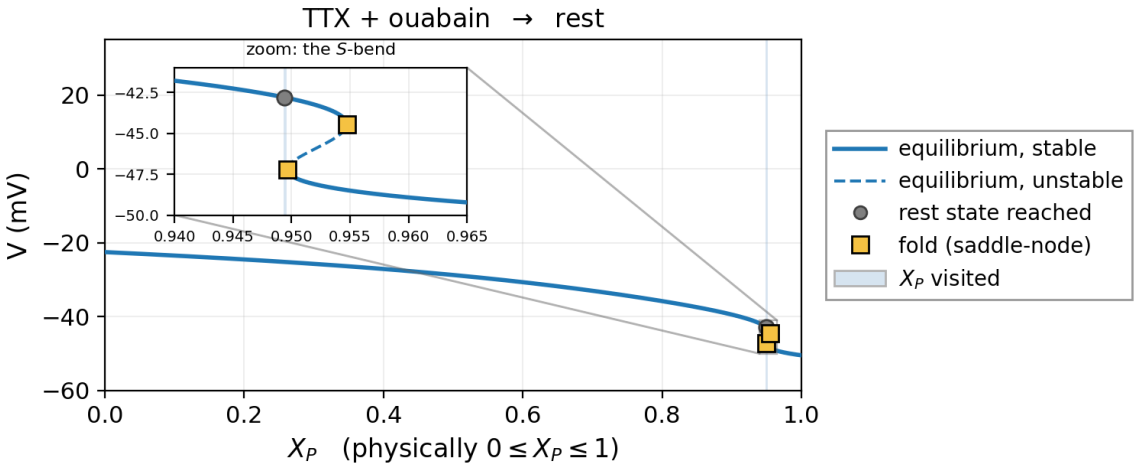


Bifurcation Analysis: TTX + Ouabain (Theodore Huxtable — SID: 540758539)

Transition from slow wave to resting behaviour as electrogenic pump is reduced under complete TTX.



Equilibrium bifurcation of ouabain under full TTX treatment



Slow variable X_p under full treatment with TTX and ouabain

Conclusion & Limitations (Emma Karpes — SID: 540695186)

The model reproduces each drug effect recorded by Junge and Stephens [1973]

	Experiment	**Our model**
Ouabain	Bursting becomes regular firing	Rest-state fold moves to $X_P = 0.96$, beyond the trajectory (≤ 0.90), so nothing ends the active phase
TTX	Spikes disappear, slow waves remain	Spiking branch destroyed ($\alpha_I \approx 0.2$); the folds remain, so the slow relaxation oscillation persists
TTX + ouabain	Slow waves suppressed	Slow wave lost at a Hopf bifurcation ($\alpha_P \approx 0.27$), leaving a stable equilibrium

Limitations

1. Thresholds (TTX spike loss, bursting-to-beating switch) found by simulation, not predicted by bifurcation analysis.
2. Slow-variable analysis explains vertical motion, not horizontal direction.
3. Hyper-polarising current, Ca-free saline and drug kinetics not modelled.

References

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Daniel T. Chiu and Richard N. Zare. Assaying for peptides in individual ¡i¿aplysia¡/i¿ neurons with mass spectrometry. *Proceedings of the National Academy of Sciences*, 95(7):3338–3340, 1998. doi: 10.1073/pnas.95.7.3338. URL https://www.pnas.org/doi/abs/10.1073/pnas.95.7.3338.

Douglas Junge and Cathy L Stephens. Cyclic variation of potassium conductance in a burst-generating neurone in aplysia. *The Journal of Physiology*, 235(1):155, 1973.

RE Plant and M Kim. Mathematical description of a bursting pacemaker neuron by a modification of the hodgkin-huxley equations. *Biophysical journal*, 16(3):227–244, 1976.